

RTP Payload Format for Windows Media Audio and Video

Version v1.0

End User License Agreement

RTP Payload Format for Windows Media Audio and Video, v1.0

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Table of Contents

1.	Introduction	1
2.	Overview	1
3.	Definition of the RTP Payload Format	3
3.1	Structure of the RTP Payload Format header	3
3.2	Grouping and fragmentation	4
3.3	Grouping of multiple fragments	6
3.4	Timing rules	7
3.4.1	Timelines	7
3.4.2	Time stamp wrap around	8
3.4.3	Decode Time and Presentation Time	8
3.4.4	Trick play	9
3.4.5	Normal Play Time	9
3.4.6	Send Time	10
3.4.7	Correspondence	11
3.5	RTP header fields	11
3.6	Payload Format header syntax	12
3.6.1	Fields in the Packet Specific Info section	12
3.6.2	Fields in the MAU Properties section	13
3.6.3	Fields in the MAU Timing section	14
3.7	Extensions	15
3.7.1	Encrypted Payload Boundary	15
3.7.2	DRM Initialization Vector	16
3.7.3	DRM Key ID	16
4.	MIME registration for Windows Media Audio	17
4.1	Required parameters	17
4.2	Optional parameters	17
4.3	SDP Parameters	18

5.	MIME registration for Windows Media Video	18
5.1	Required parameters	18
5.2	Optional parameters	19
5.3	SDP Parameters	20
6.	Security Evaluation	20
7.	References	22

1. Introduction

This specification is organized as follows:

- Introduction to this specification document.
- Section 2 provides an overview of the Real-time Transport Protocol (RTP) Payload Format specification.
- Section 3 provides the RTP Payload Format specification.
- Sections 4 and 5 define the MIME parameters and SDP syntax for content in Windows Media Audio (WMA) and Windows Media Video (WMV) formats, respectively.
- Section 6 contains a security evaluation of the RTP Payload Format.

This specification defines a RTP Payload Format for content encoded using the Windows Media Audio (WMA) or Windows Media Video (WMV) codecs. The specification also defines MIME parameters for WMA and WMV, and defines how the parameters are mapped to Session Description Protocol (SDP).

2. Overview

The RTP Payload Format defined in this document has been designed to encapsulate WMV or WMA codec bit streams. The bit streams are comprised of media access units (MAUs). This document defines how MAUs can be encapsulated in RTP.

For the purposes of this specification, a MAU is typically a single video picture or one or more WMA packets, including start codes and padding bits.

The RTP Payload Format defines the meaning of some of the fields in the RTP header. The syntax of the RTP header is defined in RFC-3550 [1]. The current specification also defines a new header, which is referred to as the RTP Payload Format header.

The RTP Payload Format header, which is defined in section 3, is inserted in front of each MAU, or fragment thereof, in the RTP packet.

Figure 1 illustrates the high-level structure of the RTP Payload Format header, and shows how it relates to the RTP header.

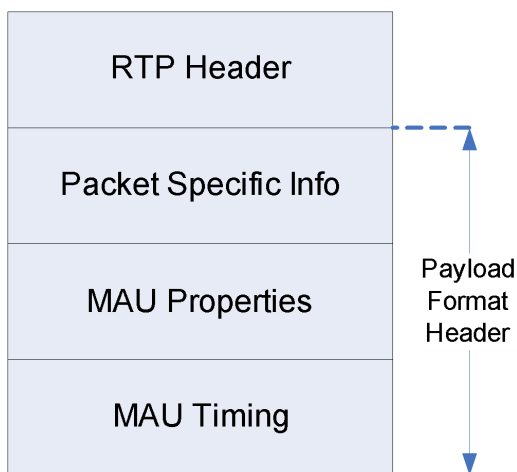


Figure 1: High-level structure of RTP Payload Format header.

As shown in Figure 1, the RTP Payload Format header is divided into three sections. Each section starts with a one-byte bit field, and is followed by one or more optional fields. In some cases, up to two entire sections may be omitted from the RTP Payload Format header. This can result in an RTP Payload Format header as small as one byte.

All RTP Payload Format fields should be transmitted in network byte order, which means that the most significant byte of each field is transmitted first.

The RTP Payload Format header is followed by a payload. The payload can consist of a complete MAU or a MAU fragment. The payload can contain a partial MAU, allowing large MAUs to be fragmented across multiple payloads in multiple RTP packets.

The first payload can be followed by additional pairs of RTP Payload Format headers and payloads, as permitted by the size of the RTP packet.

The first section of the RTP Payload Format header, which is called Packet Specific Info in Figure 1, contains information that is specific to all payloads in the RTP packet.

The Packet Specific Info section is only included once in each RTP packet, in the first RTP Payload Format header.

The second section, called MAU Properties, contains information that describes the payload that follows the header. This section specifies whether the payload contains a MAU that is a synchronization point (sync-point), such as a video key frame. It also specifies how the size of the payload is determined. Additionally, this section contains information that enables a receiver to parse the RTP packet if the previous packet was lost. This is important if a MAU is fragmented across multiple RTP packets.

The primary purpose of the third section, called MAU Timing, is to provide information about various time stamps that are associated with the MAU in the payload. This section specifies how the presentation time of the MAU is determined. This section also includes extension mechanisms that allow additional information to be included in the RTP Payload Format header.

3. Definition of the RTP Payload Format

3.1 Structure of the RTP Payload Format header

Figure 2 shows the design of the RTP Payload Format header in greater detail. Each of the three sections in Figure 1 consists of several individual header fields. These fields are shown as boxes in Figure 2.

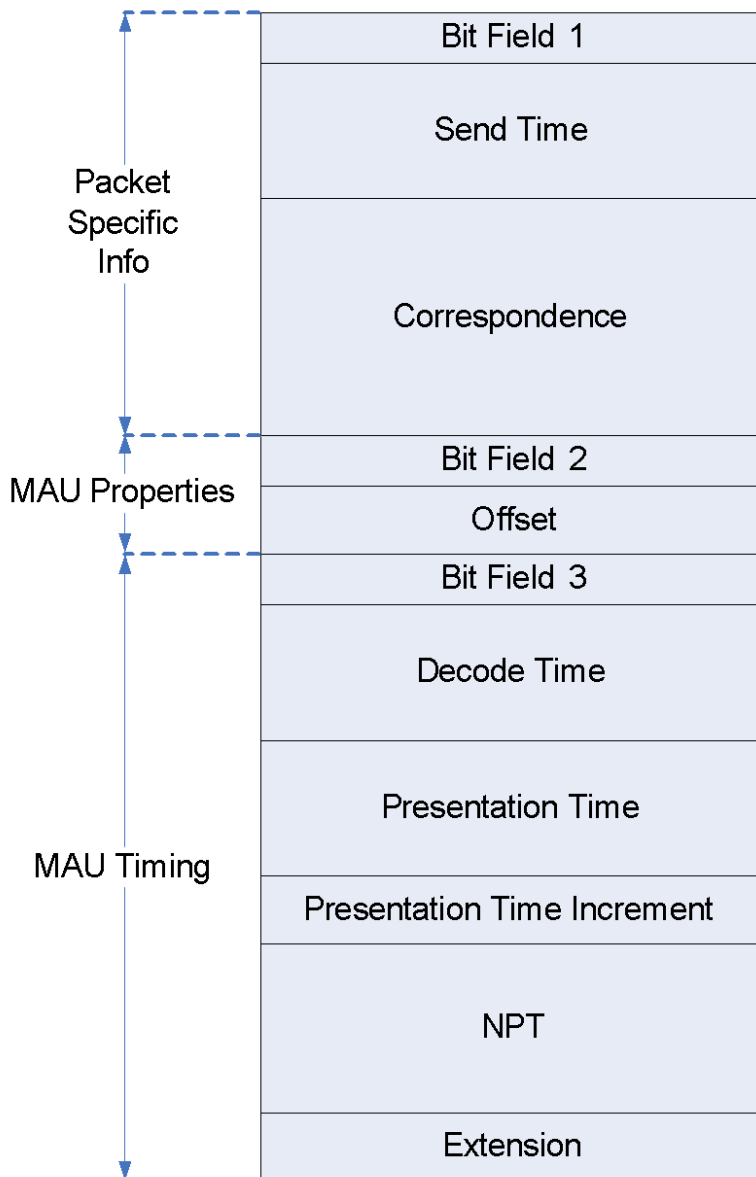


Figure 2: RTP Payload Format header.

The first header field in each of the three sections is a bit field. The other header fields in a section are present only if indicated by that section's bit field. In some cases an entire section, including its bit field, can be omitted.

The first RTP Payload Format header in the RTP packet will always include Bit Field 1, and may also include any of the other fields shown in Figure 2. Additional RTP Payload Format headers in the same RTP packet always begin with similar bit fields (Bit Field 2 and Bit Field 3 in Figure 2), and can only include the fields in the MAU Properties section and the MAU Timing section.

In the simplest possible case, an RTP packet contains a single, complete, MAU. In this case, it is possible to include all of the header fields. However, fields that are not needed can be omitted. Each of the three sections of the RTP Payload Format header has a bit field that indicates which, if any, of the fields in the section are present.

RTP Payload Format for Windows Media Audio and Video

For example, the Offset field, which specifies the byte offset to the end of the current payload, is not needed when the packet contains a single payload, because the length of the payload can be computed from the size of the RTP packet. The OP bit in Bit Field 2 indicates whether the Offset field is present. If all of the bits in Bit Field 3 are zero, then Bit Field 3 itself can be omitted, and this is indicated by setting the B3P bit in Bit Field 2 to zero.

3.2 Grouping and fragmentation

It is possible to combine multiple payloads in a single RTP packet. This is referred to as grouping. The Offset field indicates the use of grouping.

If the Offset field is present, another RTP Payload Format header and another payload may follow after the end of the current payload. The Offset field specifies the number of bytes to the end the current payload, counted from the end of the Offset field.

To determine whether another RTP Payload Format header follows the end of the current payload, consider both the value of the Offset field, the size of the RTP packet, and the size of the RTP padding area, if any.

A single MAU can be split into multiple payloads. This is referred to as fragmentation. Fragmentation is useful when a MAU is larger than a single RTP packet. The F field in Bit Field 2 indicates whether a payload contains a complete MAU or a fragment.

The fields in the MAU Timing section should only be specified in the RTP Payload Format header for the payload that contains the first fragment of a MAU. The only exception to this is when the Extension field in the MAU Timing section contains a different extension than other fragments of the same MAU. For an example of such an extension, see section 3.7.1, Encrypted Payload Boundary extension, later in this specification.

When a MAU is fragmented, the bits S, D1, and D2 in Bit Field 2 are only significant in the RTP Payload Format header for the payload that contains the first fragment.

Therefore, receivers must ignore these bits if the value of the F field is 0 or 2. For more information about the F field, see section 3.6.2 later in this specification.

Fragmentation of a MAU is not recommended unless the MAU is too large to fit in a single RTP packet. It is also generally not recommended to combine a fragment of one MAU with another MAU or a fragment of another MAU in a single RTP packet. However, receivers must still be able to handle these cases. An example of this is shown in Figure 3.

Figure 3 illustrates a possible sequence of three RTP packets. The three RTP packets transport the data of four MAUs. The remainder of the fourth MAU is continued in a fourth RTP packet (not shown). Figure 3 also illustrates that fragmented MAUs can be used to create fixed-size RTP packets.

RTP Payload Format for Windows Media Audio and Video

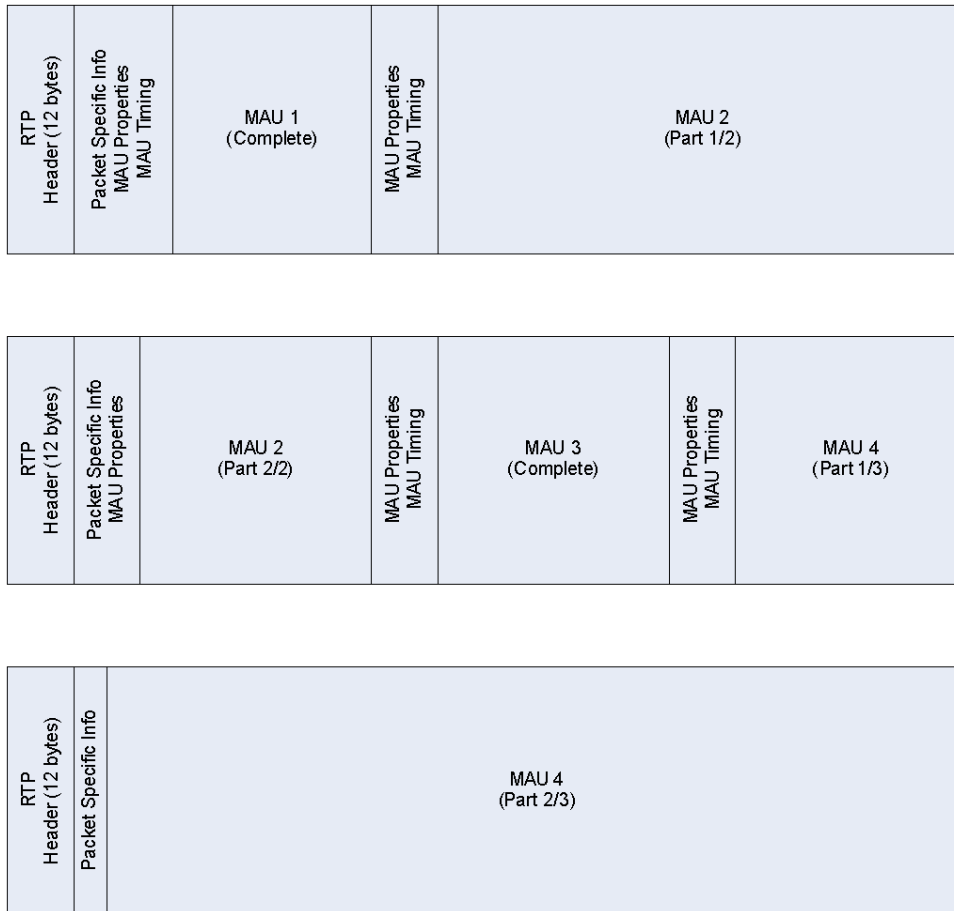


Figure 3: Example of RTP packets containing multiple MAUs.

In Figure 3, MAU 2 is fragmented across two RTP packets. In the first RTP packet, the RTP Payload Format header for MAU 2 specifies that MAU 2 is continued in the next RTP packet. (This is signaled by using the F field in Bit Field 2, which is defined in section 3.6.2.)

The second RTP packet starts with an RTP Payload Format header that does not include the MAU Timing field because the MAU Timing field for MAU 2 had already been specified in the first RTP packet. The Offset field in the MAU Properties section is used to specify the start of the RTP Payload Format header for MAU 3. This allows the client to decode MAU 3 even if the previous RTP packet was lost.

RTP Payload Format for Windows Media Audio and Video

Figure 3 also shows that MAU 4 is fragmented across the second and third RTP packets. However, MAU 4 is so large that no additional MAUs can be inserted in the third RTP packet. In this example, MAU 4 is continued in a fourth RTP packet that is not shown. In situations such as this where a MAU is very large, the third RTP packet's RTP Payload Format header does not need to include the Offset field, and it may be possible to omit the entire MAU Properties section. The remaining part of the RTP Payload Format header then consists only of the Packet Specific Info section, which can be as small as a single byte.

3.3 Grouping of multiple fragments

If a MAU is fragmented into multiple payloads, the payloads are usually carried in separate RTP packets. However, this RTP Payload Format also allows multiple payloads for the same MAU to be carried within a single RTP packet. Normally, there is no benefit in fragmenting a MAU, because extra fragmentation only increases the overhead. But this feature may be useful when the MAUs have already been fragmented without regard for the RTP packet size, and it is impractical to recombine the fragments into complete MAUs prior to RTP encapsulation.

For example, a MAU may already be fragmented when DRM-encrypted content has been stored in an ASF file [4]. In the ASF file, each MAU may be fragmented into multiple encryption units. The encryption may make it difficult for the RTP sender to recombine the encryption units into a single MAU. It may then be preferable to directly map each encryption unit to a RTP payload. If the RTP payloads are small, it may be desirable to combine several payloads into a single RTP packet.

The F field in Bit Field 2 indicates when a payload in the RTP packet contains a MAU fragment. This field is described in greater detail in section 3.6.2.

Figure 4 shows an example where a single MAU has been split into three fragments and all three fragments are carried in the same RTP packet.

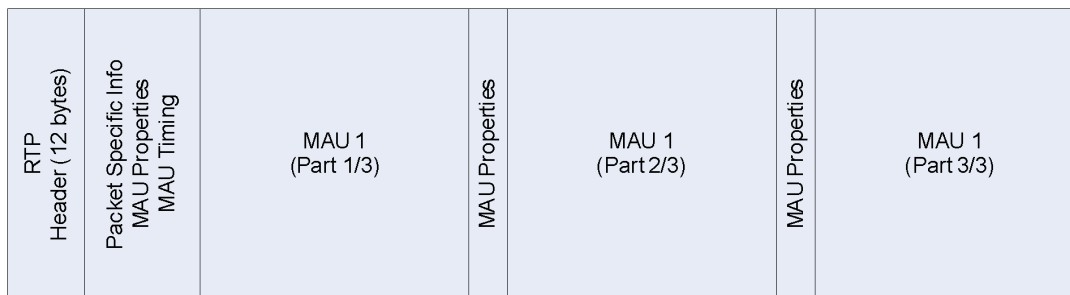


Figure 4: Example of grouping MAU fragments.

In Figure 4, the F field in the first RTP Payload Format header is set to 1, indicating that the first payload contains the first fragment of the MAU. The MAU Timing section should only be present in the first payload.

The F field in the second RTP Payload Format header is set to 0, indicating that its payload contains a fragment that is neither the first nor the last fragment of the MAU.

The F field in the third RTP Payload Format header is set to 2, indicating that its payload contains the last fragment of the MAU.

3.4 Timing rules

In addition to the usual RTP sampling clock and wallclock, this RTP Payload Format specification defines several additional time stamps and notions of time, which are described in this section.

3.4.1 Timelines

The RTP header has a single time stamp, which specifies the time at which the data in the packet was sampled. This is sometimes called the sampling clock.

It is important to note that the RTP time stamps of packets from different media streams should not be compared because the sampling clock may run at different frequencies for different media streams. For example, the sampling clock of an audio stream may run at 44,100 Hertz (Hz), while the sampling clock of a video stream will usually run at 90,000 Hz. Furthermore, RFC-3550 specifies that the value for the initial RTP time stamp should be chosen randomly. In effect, each media stream has its own timeline. In this specification, each such timeline is referred to as a media timeline.

RTP allows the timelines for different media streams to be synchronized to the timeline of a reference clock, called the wallclock. RTP senders allow the receiver to perform this synchronization by transmitting a mapping between the sampling clock and the wallclock in the Real-Time Control Protocol (RTCP) Sender Report packet. A different RTCP Sender Report must be sent for each media stream because the media streams may use different sampling clocks.

The mappings are updated and transmitted again at some interval to allow the receiver to correct for possible drift between the wallclock and the sampling clocks. Clock drift may still be a problem if the sender's wallclock drifts in relation to the receiver's wallclock. The two clocks could be synchronized by using the Network Time Protocol (NTP), for example, but the RTP specification (RFC-3550) does not specify a particular synchronization method.

It is important to note that the wallclock originates from the encoder. If the RTP sender and the encoder are separate entities, the wallclock is typically unrelated to any physical clock at the sender.

This RTP Payload Format specification uses a third timeline, called the Normal Play Time (NPT) timeline. The NPT timeline is useful primarily when RTP is used to transmit a media presentation. Time stamps from the NPT timeline usually start at 0 at the beginning of the presentation. NPT time stamps are particularly useful when transmitting a pre-recorded presentation because the time stamps can assist the receiver in specifying a playback position to seek to within the presentation. This assumes the existence of some mechanism for the receiver to communicate the new position to the RTP sender.

Since RTP was designed for digital media conferencing applications, the RTP specification does not discuss the NPT timeline. However, other protocols that are built on top of RTP include the concept of the NPT timeline. This includes protocols such as Real Time Streaming Protocol (RTSP), a control protocol for video-on-demand applications. In RTSP, the control protocol provides a mapping between the NPT timeline and the media timeline for each media stream.

This RTP Payload Format specification defines a mechanism for specifying the NPT timeline time stamp associated with a MAU. However, when practical, an out-of-band mapping between the media timeline and the NPT timeline, such as the one defined by

RTSP, may be preferable, since it reduces the overhead of the RTP Payload Format header.

3.4.2 Time stamp wrap around

All RTP-compliant systems are required to handle time stamp wraparound. At a typical clock frequency of 90,000 Hz, the RTP time stamp will wrap around approximately every 13 hours. But since the RTP specification says that a random offset should be added to the sampling clock, a receiver may experience the first wraparound in significantly less than 13 hours.

The wrapping around of the RTP time stamp is usually handled by using modular arithmetic. When modular arithmetic is used, time stamps are usually compared by subtracting one time stamp from another and observing whether the result is positive or negative. See RFC-3550 [1] for an example of how to use modular arithmetic to cope with time stamp wraparound.

It should be noted that at 90,000 Hz, 32 bits is sufficient for media timeline time stamps. Using 32 bits becomes a problem only when comparing two time stamps that are more than 6.5 hours apart.

The wallclock time is expressed by using an NTP-format time stamp. This time stamp will wrap around in the year 2036. As with the media timeline time stamps, this should not be a problem, as receivers should only need to examine the relative difference of such time stamps.

3.4.3 Decode Time and Presentation Time

In this RTP Payload Format specification, each MAU has a Decode Time and a Presentation Time. The decode time is the time by which the MAU should be delivered to the receiver's decoder, and the presentation time is the time at which the MAU should be presented (displayed or played) by the receiver. Both times belong to the media timeline, described in section 3.4.1. Since the delays in the network and in the decoder are not typically known to the RTP sender, the receiver should not use the absolute values of a decode time stamp or a presentation time stamp. The receiver should consider only the relative difference between a pair of decode time stamps or a pair of presentation time stamps.

In some cases, such as when a video codec produces bi-directional video frames, MAUs may have to be decoded in a different order than they will be presented. The RTP sender must transmit the MAUs in the order they should be decoded.

The Timestamp field in the RTP header is normally the presentation time of the first MAU in the RTP packet. This definition is equivalent to the concept of "sampling instant" in RFC-3550. Alternatively, the Timestamp field can be set to the decode time of the first MAU in the RTP packet, if this usage is negotiated through non-RTP means. Section 5.2 defines how this negotiation is done using SDP.

Since the RTP packets are transmitted in decode order, the presentation time of consecutive MAUs may not be monotonically non-decreasing.

The RTP Payload Format header includes an optional Decode Time field, which is used to specify the decode time of the MAU in the payload. The RTP Payload Format header also includes a Presentation Time field which is used to specify the presentation time of the MAU when the RTP packet contains more than one MAU.

When only a single MAU is included in the RTP packet, either the Presentation Time field or the Decode Time field is not necessary because the Timestamp field serves as a replacement for one of those fields in the first MAU in the packet.

Both the Decode Time and the Presentation Time fields are expressed using the same clock resolution as the Timestamp field.

3.4.4 Trick play

In this RTP Payload Format specification, the term "trick play" refers to the receiver rendering the media presentation at a rate other than real time. Examples of trick play include fast-forwarding and rewinding the presentation. If the RTP sender is transmitting in trick play mode, the decode time stamp and presentation time stamp for each MAU should increment at the real-time rate. This allows the decoder to decode the MAUs without knowing that trick play is used.

During trick play, the Decode Time and Presentation Time fields in the RTP Payload Format header are unaffected by trick play in that they continue to increase at their normal real-time rate. The NPT field, if present, is affected by trick play. For example, if a media presentation is being rewound, the Presentation Time time stamp fields of MAUs increase, while the value of the NPT fields decrease.

3.4.5 Normal Play Time

The NPT field in the RTP Payload Format header specifies the position in the Normal Play Time timeline to which the MAU belongs.

If the NPT field is not present, a receiver can calculate the normal play time of the MAU from the presentation time, provided that a mapping between the two timelines is available. Various approaches for establishing this mapping are discussed below.

RTP Payload Format for Windows Media Audio and Video

Since the RTP sender adds a random offset to the time stamps in the media timeline, the presentation time stamp cannot be used as a direct replacement for the NPT time stamp. Even if this random offset is known to the receiver, the wrap around of the media timeline time stamps can be an issue.

To resolve this issue, the sender can use an out-of-band mechanism to provide a mapping between the Normal Play Time timeline and the media timeline. This mapping could be provided only once, at the beginning of the transmission, or repeatedly as needed. Additionally, if trick play is possible, the sender needs to communicate the trick play rate. For example, if the presentation is being rewound, the trick play rate is negative. The receiver uses the trick play rate to generate NPT time stamps that decrease as the presentation time increases.

If the mapping is provided only once at the beginning of the transmission, the receiver should establish mapping between the Normal Play Time timeline and the wallclock timeline. This is usually possible as soon as an appropriate RTCP Sender Report packet is received. It is preferable to calculate the NPT time stamp for each MAU based on the MAU's wallclock time, because time stamps from the media timeline may drift against the wallclock timeline.

The RTSP protocol is an example of a control protocol that provides a mapping between the Normal Play Time timeline and the media timeline at the beginning of the transmission.

Another solution that may provide a suitable trade-off between complexity and overhead is to include the NPT field only on sync-point MAUs. The NPT field is used to establish a mapping between the Normal Play Time timeline and the presentation or wallclock timelines. For non-sync point MAUs, the receiver calculates the NPT time stamp using the previously established mapping. When trick play is used, the sender would include the NPT field for every MAU.

3.4.6 Send Time

The Send Time field in the RTP Payload Format header specifies the transmission time of the RTP packet. This can be useful when a sequence of RTP packets is transferred from one server to a second server. Only the first server needs to compute a transmission schedule for the packets. The second server will forward the RTP packets to other clients based on the value of the Send Time field.

It is not necessary to include the Send Time field when forwarding RTP packets to a client. However, clients can use the Send Time field to detect network congestion by comparing the difference between the values of the Send Time fields in a series of packets against the difference in packet arrival times.

The Send Time field uses the same units as the media timeline.

3.4.7 Correspondence

The purpose of the Correspondence field is to provide a mapping between the wallclock timeline and the current media timeline. This is the same mapping that is provided by RTCP Sender Reports, but the optional inclusion of this mapping in the RTP packet allows the sender to transmit the mapping more frequently than the RTCP reporting interval. Since including the mapping in the RTP packet is more efficient than transmitting a separate RTCP packet, this may allow the sender to reduce the frequency of RTCP Sender Reports, while still being able to transmit the mapping as frequently as desired.

3.5 RTP header fields

Figure 5 illustrates the standard 12-byte RTP header. It is the same header that is defined in RFC-3550 [1], and it is shown here for reference only.

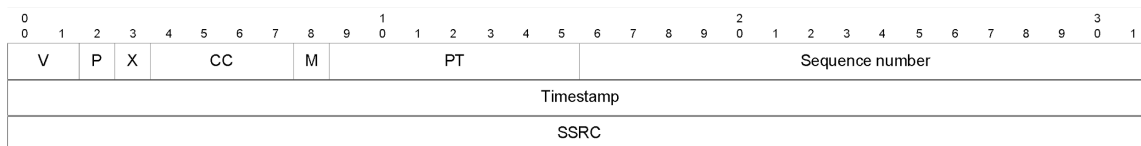


Figure 5: Standard 12-byte RTP header.

Explanation of header fields:

Version (V) field. 2 bits. This field must always be set 2.

Padding (P) bit. This bit is used to add padding to the end of the RTP packet, as specified in RFC-3550.

Extension (X) bit. This bit is set to 1 if an RTP header extension is present. The RTP profile defines how the header extension is used. A receiver must be able to parse, or skip over, the header extension should the RTP header have a non-zero Extension bit.

Contributing Source (CC) field. 4 bits. This field is used as defined in RFC-3550. A receiver must be able to correctly parse, or skip over, the list of contributing sources should the RTP header have a non-zero Contributing Source field.

Marker (M) bit. This bit is set to 1 if any of the payloads in the RTP packet contain complete a MAU, or the last fragment of a MAU.

Payload Type (PT) field. 7 bits. The assignment of an RTP payload type is beyond the scope of this specification. It is specified by the RTP profile under which this payload format is used, or signaled dynamically out-of-band (such as by using SDP).

Sequence number field. 16 bits. This field contains a number that increments by 1 for each RTP packet that is sent with the same SSRC value. The initial value of the RTP sequence number may be communicated to the client through non-RTP means.

Timestamp field. 32 bits. This field specifies a time stamp that applies to the first payload that is included in the RTP packet. By default, the field is interpreted as a presentation time. Alternatively, the Timestamp field can be interpreted as a decode time, if this usage is negotiated through non-RTP means. The clock frequency of the Timestamp field is recommended to be 90 kilohertz (kHz), i.e., the resolution is 1/90,000 seconds. The sender and receiver may negotiate a different clock frequency through non-RTP means.

Synchronization Source (SSRC) field. 32 bits. RTP packets with the same value for the SSRC field share the same timeline for the Timestamp field and the same number space for the Sequence Number field.

3.6 Payload Format header syntax

The RTP header is always followed by a RTP Payload Format header. The only exception is a RTP packet that only consists of padding. In that case, the RTP Payload Format header is not present. If a RTP packet contains data from multiple MAUs, the RTP Payload Format header will appear in front of each MAU. Thus, RTP packets using this payload format may contain one or more RTP payload Format headers.

The layout of the RTP Payload Format header is shown in Figure 2. When the RTP Payload Format header directly follows the standard 12-byte RTP header, it must begin with the one-byte field called Bit Field 1, followed by a series of optional fields. The header is followed by a payload. The payload consists of either a complete MAU or a fragment (partial) MAU.

After the first data payload, another RTP Payload Format header may appear, followed by another data payload. The process of adding another RTP Payload Format header after a data payload may be repeated multiple times. Each RTP Payload Format header that follows the first data payload must begin with the Bit field 2 field.

3.6.1 Fields in the Packet Specific Info section

The following diagram shows the layout of the field Bit Field 1.

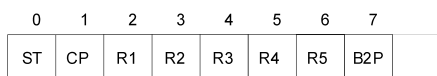


Figure 6: Layout of Bit Field 1.

Send Time Present (ST) bit. If this bit is 1, a 32-bit Send Time field is inserted directly following the end of the Bit Field 1 field.

Correspondence Present (CP) bit. If this bit is 1, a 96-bit Correspondence field is inserted after the Send Time field.

R1, R2, and R3 fields. 1 bit each. If the bit is set to 1, the receiver should assume that a 32-bit field has been added between the Correspondence field and Bit Field 2. The meaning of these 32-bit fields is not defined in this specification. A receiver that does not understand the 32-bit fields must ignore them.

R4 and R5 fields. 1 bit each. Reserved for future use and must be set to 0.

Bit Field 2 Present (B2P) bit. If this bit is 1, the one-byte Bit Field 2 field is inserted after the Correspondence field.

Send Time field. 32 bits. This field specifies the transmission time of the RTP packet, using the same time units that are used for the Timestamp field in the RTP header.

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Correspondence field. 96 bits. Consists of two time stamps: A 64-bit wallclock time stamp in NTP format and a 32-bit decode time stamp. The two fields are used in the same way as the NTP time stamp and the RTP time stamp in the RTCP Sender Report, which is defined in section 6.4.1 of RFC-3550.

3.6.2 Fields in the MAU Properties section

When Bit Field 1 is present, Bit Field 2 is optional. The B2P bit in Bit Field 1 determines whether Bit Field 2 is present. The default value for all bits in Bit Field 2 is 0.

0	1	2	3	4	5	6	7
F	OP	S	D1	D2	E	B3P	

Figure 7: Layout of Bit Field 2.

Fragmentation (F) field. This field indicates whether the data payload consists of a partial MAU. One or more payloads must be combined to reconstruct a complete MAU. The F field also indicates whether the payload contains the first or last fragment of the MAU.

Table 1: Meaning of F field.

F field value	Meaning
0	Payload contains fragment of MAU other than the first or last fragment.
1	Payload contains first fragment of MAU.
2	Payload contains last fragment of MAU.
3	Payload contains complete MAU (not fragmented.)

The S, D1 and D2 bits are only valid when the value of the F field is 0 or 3.

Offset Present (OP) bit. If this bit is 1, the 16-bit Offset field is inserted directly after Bit Field 2. The Offset field is used to find the end of the current payload. Another RTP Payload Format header, starting with Bit Field 2, may follow the end of the current payload. If the Offset Present bit is 0, the Offset field is absent, and the current payload extends to the end of the RTP packet or to the start of the RTP padding area, if the Padding bit in the RTP header is 1.

Sync Point (S) bit. This bit is set to 1 when the MAU is a sync-point MAU.

Discontinuity (D1) bit. This bit is set to 1 to indicate that one or more MAUs are missing even though RTP sequence number does not have a gap.

Droppable (D2) bit. If this bit is 1 and it is necessary to drop some MAUs, this MAU can be dropped with less negative impact than MAUs that have the D2 bit set to 0.

Encryption (E) bit. Set to 1 to indicate that the payload contains encrypted data. The bit should be set to 0 if the payload does not contain encrypted data.

Bit Field 3 Present (B3P) bit. If this bit is 1, the one-byte Bit Field 3 field is inserted after the Length field.

Offset. A 16-bit field that specifies the offset, in bytes, to the end of the current payload, counted from the first byte following the Offset field. In other words, the value of the Offset field is the size of the MAU Timing section, if any, plus the size of the current payload.

3.6.3 Fields in the MAU Timing section

The value of the B3P bit in Bit Field 2 determines whether Bit Field 3 is present. The default value for all bits in Bit Field 3 is 0.

0	1	2	3	4	5	6	7
D3	P	N	R6	R7	R8	R9	X

Figure 8: Layout of Bit Field 3.

Decode Time Present (D3) bit. If this bit is 1, the 32-bit Decode Time field is inserted after Bit Field 3, but before the Presentation Time field.

Presentation Time Present (P) bit. If this bit is 1, the 32-bit Presentation Time field is inserted after the Decode Time field, but before the NPT field.

NPT Present (N) bit. If this bit is 1, the 64-bit NPT field is inserted immediately after the Presentation Time field.

R6, R7, R8, and R9 fields. For each of these bits that are set to 1, the receiver should assume that a 32-bit field has been added between the NPT field and the Extension field. The meaning of these 32-bit fields is not defined in this specification. A receiver that does not know the meaning of the 32-bit fields must ignore them.

Extension Present (X) bit. If this bit is 1, a variable-size Extension field is inserted after the NPT field.

Decode Time. A 32-bit field that specifies the decode time of the MAU, using the same time units that are used for the Timestamp field in the RTP header.

Presentation Time. A 32-bit field that specifies the presentation time of the MAU, using the same time units that are used for the Timestamp field in the RTP header.

NPT field. A 64-bit time stamp using the same time units as the RTP Timestamp field. The NPT field specifies the position in the Normal Play Time timeline to which the MAU belongs. See sections 3.4.4 and 3.4.5 for a description of how this field is used.

Extension field. Shown in Figure 2, this field consists of one or more collections of fields. Figure 9 illustrates the layout of the fields contained in one such collection.



Figure 9: Layout of Extension field.

L bit. If this bit is 1, this is the last collection of Extension fields. If the bit is 0, the end of the Extension Data field is followed by at least one more collection of Extension fields as shown in Figure 9.

Extension Type. A 7-bit field that is used to identify the contents of the Extension Data field. Section 3.7 lists the currently defined values for the Extension Type field. The values 0 and 127 are reserved for future use.

Extension Length. An 8-bit number giving the size, in bytes, of the Extension Data field that appears directly following this field.

Extension Data. Variable-length field. The size of this field is given by the Extension Length field.

3.7 Extensions

This section defines extensions that can be used in the Extension field.

3.7.1 Encrypted Payload Boundary

The fields in the Extension field have the following values when the Encrypted Payload Boundary extension is used.

Extension Type. Must be 1.

Extension Length. Must be 1.

Extension Data. One byte. The most significant bit is referred to as the G bit. All other bits must be set to zero by the sender and must be ignored by receivers.

The default value of the G bit is zero. Since no other bits are used in this extension, this extension should not be included when the value of G is zero.

The G bit is used to indicate the boundaries of DRM encryption units. When certain versions of DRM are used, a single MAU may be divided into multiple encryption units. Each such encryption unit is transmitted as a separate payload. If an encryption unit must be fragmented into multiple payloads, the G bit is set to 1 on all fragments except the last one.

If the G bit is 1 and DRM is used, the receiver must reassemble the payloads into a complete encryption unit before it can decrypt it.

If the G bit is 0 and DRM is used, the payload contains either a complete encryption unit or the last fragment of a sequence of fragmented encryption units.

The G bit should always be 0 when DRM is not used.

3.7.2 DRM Initialization Vector

The fields in the Extension field have the following values when the DRM Initialization Vector extension is used.

Extension Type. Must be 2.

Extension Length. The size of the Extension Data field, in bytes.

Extension Data. A sequence of one or more bytes, to be used as part of the initialization vector for the current MAU. When this extension is present, the DRM encryption unit is always a complete MAU. If the MAU is fragmented into multiple payloads, the DRM Initialization Vector extension should be present only in the first payload.

The Encrypted Payload Boundary extension, defined in section 3.7.1, must not be used when this extension is used.

3.7.3 DRM Key ID

The fields in the Extension field have the following values when the DRM Key ID extension is used.

Extension Type. Must be 3.

Extension Length. The size of the Extension Data field, in bytes.

Extension Data. A sequence of one or more bytes that identifies the decryption key to use for decrypting the current payload.

The DRM Key ID extension remains effective until replaced by a different DRM Key ID extension. Therefore, the extension is only required when a payload requires the use of a decryption key that is different from the decryption key of the previous payload. However, if the previous payload was contained in an RTP packet that was lost, the receiver may be unaware that a change of decryption key is necessary. If a payload is decrypted with the wrong key, it can lead to undesirable rendering artifacts.

One approach to reducing the severity of this issue is to specify the DRM Key ID extension for the first payload of every MAU that is a sync-point. This is a good solution if the lost MAU will force the receiver to discard all MAUs until it receives the next sync-point MAU. A more conservative solution is to specify the DRM Key ID extension for the first payload in each multiple-payload RTP packet. This solution guards against packet loss because the interdependent payloads are all contained within a single RTP packet.

4. MIME registration for Windows Media Audio

The receiver must ignore any unspecified parameters.

Media Type name: audio

Media subtype name: wma

4.1 Required parameters

version. A case-insensitive string that specifies the version of the Windows Media Audio codec. Table 2 shows the currently defined values of the version parameter:

Table 2: Values for WMA version parameter.

Version parameter	Description
STD	Windows Media Audio 9 Standard
PRO	Windows Media Audio 9 Professional
LSL	Windows Media Audio 9 Lossless
VOICE	Windows Media Audio 9 Voice

bitrate. The average bit rate, in bits per second, of the encoded audio stream. The effect of RTP encapsulation is not included in this number.

blocksize. The maximum size, in bytes, of the WMA audio packets. This number is independent of the RTP packet size, as multiple WMA audio packets can be combined in a single RTP packet.

samplesize. The number of bits per audio sample.

preroll. The number of milliseconds worth of data that must be received before decoding can begin to ensure smooth playback of the decoded audio stream.

config. A Base16 [3] (hexadecimal) representation of the WMA Format Header. The bits in the Format Header are represented with the most significant bit first. If the Format Header is not a multiple of 8 bits, up to 7 zero-valued padding bits should be added after the last bit, to extend the Format Header to a multiple of 8 bits.

4.2 Optional parameters

profile. A number that specifies the encoding profile used in the audio stream.

samplerate. The number of samples per second in the encoded audio stream. This parameter is optional if the a=rtpmap line in the Session Description Protocol (SDP) specifies the sample rate. The parameter must be specified if a=rtpmap specifies an RTP clock frequency that is different from the sample rate of the encoded audio stream.

4.3 SDP Parameters

The MIME media type audio/wma is mapped to fields in the SDP as follows:

- The media name in the m= line of SDP must be audio.
- The encoding name in the a=rtpmap line of SDP must be WMA (the MIME subtype).
- The clock frequency in the a=rtpmap line should be equal to the audio sample rate.

RTP Payload Format for Windows Media Audio and Video

- The encoding parameters section in the a=rtpmap line must be set to the number of channels in the audio stream.
- The optional and required parameters, described above, must be included in the a=fmtp line of SDP. These parameters are expressed as a semicolon separated list of name-value pairs. Names are separated from their corresponding values by a = character.

Example:

```
m=audio 50000 RTP/AVP 98
a=rtpmap:98 wma/48000/2
a=fmtp:98 version=STD;profile=2;bitrate=64000;blocksize=8000;
    samplesize=16;preroll=3000;config=ABCDEF
```

5. MIME registration for Windows Media Video

The receiver must ignore any unspecified parameters.

Media Type name: video

Media subtype name: wmv

5.1 Required parameters

version. A number that specifies the version of the Windows Media Video codec. Table 3 describes the currently defined values for this parameter.

Table 3: Values for WMV version parameter.

Version parameter	Description
WMV1	Windows Media Video 7
WMV2	Windows Media Video 8
WMV3	Windows Media Video 9
WMVA	Windows Media Video Advanced Profile

bitrate. The average bit rate, in bits per second, of the encoded video stream. The effect of RTP encapsulation is not included in this number.

width. The number of horizontal pixels in the encoded video image.

height. The number of vertical pixels in the encoded video image.

preroll. The number of milliseconds worth of data that must be received before decoding can begin in order to ensure smooth playback of the decoded video images.

config. A Base16 [3] (hexadecimal) representation of the WMV Sequence Header. The bits in the Sequence Header are represented with the most significant bit first. If the Sequence Header is not a multiple of 8 bits, up to 7 zero-valued padding bits should be added after the last bit, to extend the Sequence Header to a multiple of 8 bits.

5.2 Optional parameters

profile. A number that specifies the encoding profile used in the video stream. This parameter should be present if the content is encoded using either the Windows Media

Video 9 or Windows Media Video Advanced Profile codec. The concept of an encoding profile is not applicable to content encoded using the Windows Media Video 7 and Windows Media Video 8 codecs. The profile parameter should be omitted when those codec versions are used. Table 4 lists the currently defined profiles.

Table 4: Values for WMV profile parameter.

Profile parameter	Description
0	Simple Profile
1	Main Profile
3	Advanced Profile

level. The encoding level used in the video stream. This parameter should be present if the content is encoded using either the Windows Media Video 9 or Windows Media Video Advanced Profile codec. The concept of an encoding level is not applicable to WMV content encoded using the Windows Media Video 7 and Windows Media Video 8 codecs. The level parameter should be omitted when those codec versions are used. The interpretation of the level parameter depends on the value of the profile parameter. Table 5 lists the currently defined levels.

Table 5: Values for WMV level parameter.

Level parameter	Simple Profile	Main Profile	Advanced Profile
1	Low Level	Low Level	Level 1
2	Medium Level	Medium Level	Level 2
3		High Level	Level 3

aspect. The aspect ratio of the encoded video image. The aspect ratio is represented as two decimal numbers, separated by a colon (:). The first number is the aspect width, and the second number is the aspect height. If this parameter is not specified, the aspect ratio is 1:1.

RTP Payload Format for Windows Media Audio and Video

ts. A string that defines the meaning of the RTP Timestamp field. The possible values for this parameter are PTS and DTS. The default is PTS. If the value of the ts parameter is PTS, the RTP Timestamp field specifies the presentation time of the first MAU in the RTP packet. If the value is DTS, the RTP Timestamp field specifies the decode time of the first MAU in the RTP packet.

5.3 SDP parameters

The MIME media type video/wmv is mapped to fields in the Session Description Protocol (SDP) as follows:

- The media name in the m= line of SDP must be video.
- The encoding name in the a=rtpmap line of SDP must be WMV (the MIME subtype).
- The clock frequency in the a=rtpmap line should be 90000.
- The optional and required parameters, described above, must be included in the a=fmtp line of SDP. These parameters are expressed as a semi-colon separated list of name-value pairs. Names are separated from their corresponding values by a = character.
- The a=framerate line [2] must be present. It is used to specify the frame rate of the video stream.

Example:

```
m=video 50000 RTP/AVP 98
a=rtpmap:98 wmv/90000
a=fmtp:98 version=WMV3;profile=2;level=3;ts=DTS;bitrate=300000;
width=320;height=240;aspect=16:11;preroll=3000;config=ABCDEF
a=framerate:24.976
```

6. Security Evaluation

Implementers should be aware that the Offset field in the RTP Payload Format header could have an invalid value. An invalid value could cause an implementation to attempt to read from unallocated memory if this situation is not handled correctly. The issue also exists with the Extension Length header.

This RTP Payload Format specification defines two fragmentation mechanisms. The first mechanism allows a sender to fragment a MAU across multiple payloads (section 3.2.) Receivers can reassemble such payloads to reconstruct the original MAU.

The second fragmentation mechanism allows payloads containing DRM-encrypted data to be fragmented into multiple payloads (section 3.7.1). Receivers can reassemble such fragmented payloads to reconstruct the original payload.

RTP Payload Format for Windows Media Audio and Video

Implementers should take note that both fragmentation mechanisms are susceptible to an attack where the attacker deliberately refrains from sending an RTP packet containing the last fragment. Unless properly mitigated, this could cause the receiver to tie up memory resources for an indeterminate amount of time.

An attacker could send a sequence of RTP packets with identical time stamps, or time stamps that decrease instead of increase. Unless properly mitigated, this could cause a receiver to buffer packets and tie up memory resources for an indeterminate amount of time.

Receivers should also be prepared for the possibility that the data contained in the MAU may have been deliberately corrupted by an attacker to cause a malfunction of the decoder.

RTP and RTCP are subject to spoofing. Spoofed RTP packets can cause the receiver to process the wrong data, possibly leading to decoder malfunction. Spoofed RTCP Sender Report packets can cause synchronization problems, and can cause the receiver to send RTCP Receiver Reports less frequently than required. Spoofed RTCP Receiver Report packets can cause the server to retransmit RTP packets that were not lost, possibly flooding the receiver with data. However, these problems are inherent to RTP and RTCP and are not introduced by this specification.

7. References

1. "[RTP: A Transport Protocol for Real-Time Applications](http://www.ietf.org/rfc/rfc3550.txt)", 2003, RFC-3550
(<http://www.ietf.org/rfc/rfc3550.txt>)
2. "[SDP: Session Description Protocol](http://www.ietf.org/rfc/rfc2327.txt)", RFC-2327 (<http://www.ietf.org/rfc/rfc2327.txt>)
3. "[The Base16, Base32 and Base64 Data Encodings](http://www.ietf.org/rfc/rfc3548.txt)", RFC-3548
(<http://www.ietf.org/rfc/rfc3548.txt>)
4. "[Advanced Systems Format \(ASF\) Specification](http://www.microsoft.com/windows/windowsmedia/format/asfspec.aspx)"
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